BEiT: BERT Pre-Training of Image Transformers

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Problem

Vision Transformers need large-scale **labeled** data — yet a simple **self-supervised** recipe was missing.

GAP

Raw-pixel masking has no ready *vocabulary* to predict.

Motivation

- Can **masked image modeling** pre-train ViTs label-free, as BERT did for language?
- Contrastive / pixel-regression SSL dominate vision; a **discrete-token** target mirrors BERT directly.

Contribution

1. A **masked image modeling (MIM)** task that predicts discrete visual tokens for masked patches.
2. **BEiT**, a self-supervised ViT that, once pre-trained, fine-tunes competitively on both classification and segmentation.
3. Analysis: the learned **self-attention separates objects** with no annotation.
4. Adds **blockwise masking** and a variational-autoencoder view linking MIM to the tokenizer.

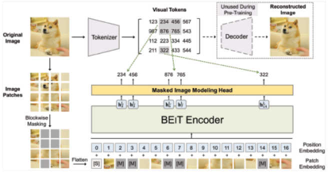
Method

1. **Two views** per image: 16×16 **patches** + discrete **visual tokens** from a pre-trained dVAE tokenizer (DALL-E, 8192-token vocabulary).
2. **Blockwise-mask ~40%** of patches and replace them with a learnable `[M]` embedding.
3. Feed the corrupted patch sequence to a **ViT** backbone; at each masked position **predict the original visual token** (MIM).
4. Append task layers and **fine-tune end-to-end** for classification or segmentation.

$$\max \sum_{x\in\mathcal{D}} \mathbb{E}_{\mathcal{M}}\left[\sum_{i\in\mathcal{M}} \log p_{\mathrm{MIM}}(z_i \mid x^{\mathcal{M}})\right]$$

recover masked visual tokens z_i from the corrupted image $x^{\mathcal{M}}$

Small-range initialization with per-layer rescaling stabilizes large-scale Transformer pre-training. MIM can be read as one stage of a **variational autoencoder**: the frozen dVAE encodes visual tokens, and the ViT learns a masked prior over them.



Overview of BEiT pre-training: an image tokenizer maps patches to discrete visual tokens; masked patches feed a ViT that predicts the tokens of the original image.

Dataset / Benchmark

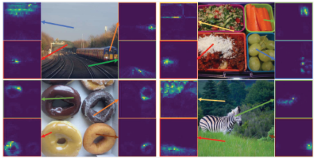
- **Pre-train:** ImageNet-1K, **1.2M images, no labels** — 800 epochs (~500k steps), 16× V100 32GB, ~5 days.
- **Evaluate** by fine-tuning on ImageNet-1K classification & ADE20K semantic segmentation.

ImageNet-1K · ADE20K · dVAE tokenizer · 8192 visual tokens

Key Results

BEiT — self-supervised, fine-tuned	ImageNet top-1
BEiT-B (base)	83.2%
BEiT-L (large)	85.2%
BEiT-L @ 384²	**86.3%**

45.6 ADE20K mIoU · **47.7** + interm. FT · **86.3%** IN @384²



Self-attention maps: BEiT separates objects with no manual annotation — semantics emerge from self-supervised pre-training.

SO WHAT → *BEiT$_{384}$ (ImageNet-1K only) surpasses supervised ViT$_{384}$ pre-trained on ImageNet-22K, and beats DINO / MoCo v3.*

Ablation Study

PRE-TRAINING VARIANT	IN TOP-1	ADE MIOU
Raw-pixel target	81.04	41.38
No blockwise mask	82.77	42.93
Full BEiT (visual tokens)	**82.86**	**44.65**

SO WHAT → ***Visual tokens** beat raw-pixel targets (83.2 / 45.6).*

Headline Numbers

83.2%
ImageNet-1K top-1 · BEiT-B

85.2% BEiT-L top-1 · **45.6** ADE20K mIoU · **~40%** patches masked

Takeaway

Predicting **discrete visual tokens** for masked patches makes BERT-style pre-training work for vision Transformers.

The "BERT moment" for vision: predict tokens, not pixels.

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